Chapter 5: Dictionary & Sets.

Dictionary is a collection of key-value pairs

Sintax :
```
a = {"key" : "value",
     "harry" : "code",
     "marks" : "100",
     "list" : [1, 2, 9] }
```

a ["key"] => Prints "value"
a ["list"] => Prints [1, 2, 9]

Properties of Python Dictionaries.

1. It is unordered
2. It is mutable
3. It is indexed
4. Cannot Contain duplicate keys.

Dictionary Methodes

Consider the following dictionary.

```
a = { "name" : "Harry",
      "from" : "India",
      "marks" : [92, 98, 96] }
```

1. a.items(): returns a list of (key, value) tup
2. a.keys(): returns a list ~~Contain~~ Containing dictionary's keys.
3. a.update({"friend":"sam"}): updates the dictionar with supplied key-value pairs.

Difference b/w a.get ("name") & a["name"]

if this key doesnt exist in the dictionary a["name"] gives error while as: a.get ["name"] returns 'none' if ~~ent~~ key not present in dict

4> a.get ("name"): returns the value of the specified keys (and value is returned eg. "Harry" is returned ~~tra~~ here).

More methods are available on docs: python.org.

Sets in Python.
Sets is a collection of non repetitive elements.

```
S = Set()
S.add (1)
S.add (2)
```

This is an empty set

=> No repatition allowed!

=> or set = {1, 2}

If you are a programming beginner without much knowledge of mathamatical operations on sets, you can simply look at sets in python as data types containing unique values.

```
S = {5,3,2} # This denots set
e = Set() # this denotes emptyset
#dont use Set = {} as it will
create empty dictionary
```

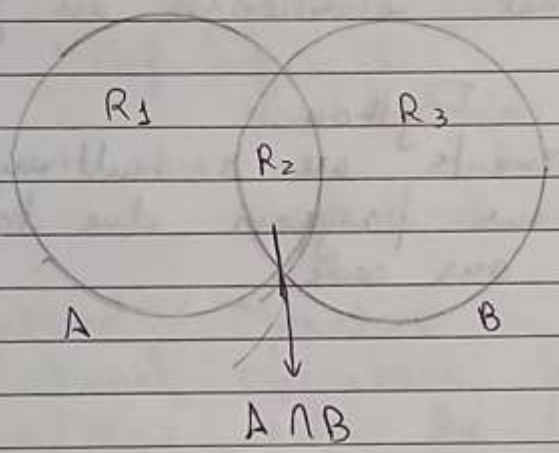
Properties of Sets.

1> Sets are unordered => Element's order doesnt matter.
2> Sets are unindexed => Cannot access elements by index.
3> There is no way to change items in sets.
4> Sets Cannot Contain duplicate values.

Operations on Sets
Consider the following set:

S = {1, 8, 2, 3}

1> Len (s) : Returns 4, the length of the set.
2> S.remove (8) : Updates the set S and removes 8 from S.
3> S.pop () : Remove an arbitrary element from the set and returns the element removed.
4> S.clear () : Empties the set S.
5> S.union ({8, 11}) : Returns a new set with all items from both sets. => {1, 8, 2, 3, 11}
6> S.intersection ({8, 11}) : Returns a set which contains only items in both sets. => {8}



$R_2 \Rightarrow A \cap B$
$R_1 + R_2 + R_3 \Rightarrow A \cup B$
$R_1 + R_3 \Rightarrow A \Delta B$
$R_1 \Rightarrow A - B$
$R_3 \Rightarrow B - A$

~~Symb~~ Symbols :

$\cap$ => intersection
$\cup$ => Union
Δ => Symmetric difference => $R_1 - R_2$ & $R_3 - R_2$
or $R_1 + R_3$